

HILO INTL, HI, USA

WMO: 912850

Lat: 19.719N

Lon: 155.053W

Elev: 12

StdP: 101.19

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 21504

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.5	17.2	10.8	8.0	18.3	11.9	8.7	18.9	10.7	23.0	9.1	23.3	2.7	220	0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	30.2	23.7	29.6	23.4	29.0	23.1	25.0	28.3	24.5	27.8	24.1	27.4	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.0	26.6	23.5	18.3	26.2	23.0	17.7	25.8	76.0	28.3	73.8	27.7	72.0	27.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.6	5.7	14.9	31.3	0.8	1.2	14.4	32.1	13.9	32.8	13.4	33.4	12.9	34.3		
(5)			11.6	25.5	0.8	1.0	11.1	26.3	10.6	26.9	10.2	27.4	9.6	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	22.0	21.9	22.3	22.6	23.4	24.0	24.7	24.9	24.8	24.4	23.4	22.4
	DBStd	1.54	1.12	1.28	1.22	1.14	1.02	0.87	0.99	0.97	1.15	1.06	1.15	1.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	4898	373	334	380	378	417	422	455	462	444	446	402	386
	CDD18.3	1857	114	101	122	128	158	172	197	203	194	188	152	127
	CDH23.3	9679	455	385	454	503	811	979	1263	1350	1292	1069	664	454
CDH26.7	1553	43	38	41	37	100	140	245	283	301	209	84	32	
Wind	WSAvg	2.9	2.9	3.1	3.1	3.0	2.9	2.8	2.8	2.8	2.8	2.7	2.7	2.8
Precipitation	PrecAvg	3187	233	264	332	311	216	173	240	266	221	253	374	304
	PrecMax	5367	819	1157	1268	1098	635	577	726	1241	636	583	1166	895
	PrecMin	1582	3	13	22	25	23	46	90	68	40	61	26	7
	PrecStd	757	204	246	266	216	138	91	127	189	124	122	247	201
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	29.3	29.0	28.6	29.5	29.6	30.2	30.6	31.2	30.5	29.7	28.6
		MCWB	21.4	21.6	22.2	22.0	22.5	22.8	23.8	24.4	24.6	23.8	23.3	22.3
	2%	DB	27.9	27.8	27.8	27.7	28.5	28.8	29.4	29.7	30.1	29.5	28.5	27.5
		MCWB	21.3	21.4	21.7	21.7	22.3	22.7	23.5	23.8	23.8	23.5	23.1	21.9
	5%	DB	26.8	26.8	26.8	27.0	27.8	28.1	28.8	29.0	29.3	28.7	27.6	26.7
		MCWB	21.0	21.0	21.2	21.4	22.0	22.5	23.2	23.5	23.4	23.2	22.6	21.4
	10%	DB	25.9	25.8	25.9	26.2	27.1	27.5	28.1	28.4	28.5	27.9	26.8	25.9
		MCWB	20.6	20.5	20.8	21.0	21.7	22.2	22.9	23.2	23.1	22.9	22.1	21.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	23.0	23.1	23.5	24.0	24.2	25.3	25.7	25.7	25.2	24.5	23.6
		MCDB	26.6	26.7	26.9	26.8	27.6	27.7	28.4	28.9	29.3	28.5	27.9	26.8
	2%	WB	22.3	22.4	22.5	22.6	23.1	23.6	24.6	24.9	24.9	24.4	23.8	22.8
		MCDB	26.0	25.9	26.1	25.9	26.9	27.2	27.8	27.9	28.4	27.9	27.1	25.9
	5%	WB	21.7	21.8	21.9	22.0	22.7	23.0	24.0	24.3	24.3	23.9	23.2	22.2
		MCDB	25.3	25.2	25.5	25.4	26.4	26.7	27.3	27.4	27.7	27.3	26.3	25.2
	10%	WB	21.1	21.2	21.3	21.5	22.2	22.7	23.5	23.8	23.7	23.4	22.7	21.7
		MCDB	24.6	24.6	24.7	24.9	25.8	26.3	26.8	27.0	27.2	26.7	25.6	24.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.8	7.4	7.1	7.4	7.3	7.3	7.1	7.6	7.4	7.1	7.3
		MCDBR	9.7	9.4	8.8	8.2	8.3	7.9	7.9	7.9	8.4	8.3	8.2	8.4
		MCWBR	6.2	5.8	5.3	4.4	4.6	4.2	4.1	4.4	4.8	4.9	5.1	5.2
	5% WB	MCDBR	8.4	8.3	8.0	7.4	7.6	7.3	7.1	7.0	7.5	7.5	7.2	7.3
		MCWBR	5.8	5.4	5.1	4.3	4.4	4.1	4.0	4.3	4.6	4.7	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.369	0.371	0.389	0.433	0.408	0.396	0.403	0.388	0.385	0.382	0.379	0.362	
	taud	2.558	2.553	2.498	2.332	2.446	2.471	2.456	2.513	2.517	2.554	2.564	2.600	
	Ebn at Noon	911	928	919	876	888	890	884	905	909	905	894	907	
	Edh at Noon	95	101	110	131	116	112	114	108	106	99	94	89	
All-Sky Solar Radiation	RadAvg	4.49	4.88	5.36	5.93	6.24	6.40	6.50	6.31	5.89	5.13	4.40	4.07	
	RadStd	0.27	0.32	0.58	0.40	0.38	0.23	0.28	0.30	0.24	0.20	0.24	0.38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+125

Nomenclature: See separate page